Hill Cipher

The Hill cipher is a poly graphic substitution cipher that uses linear algebra concepts to encrypt messages. It encrypts blocks of text using a key matrix and operates on the principles of matrix multiplication and modular arithmetic. Here's a detailed breakdown of the Hill cipher, including its encryption and decryption processes.

Key Concepts

1. **Key Matrix**: The key must be a square matrix (e.g., 2×2, 3×3) that is invertible in modular arithmetic (specifically mod 26 for the English alphabet).
2. **Plaintext Representation**: The plaintext is converted into numerical values (A=0, B=1, ..., Z=25) and grouped into blocks that match the size of the key matrix.
3. **Modular Arithmetic**: All calculations are performed modulo 26 to handle wraparounds in the alphabet.

Steps to Encrypt Using the Hill Cipher

1. **Choose a Key Matrix**: Select a square matrix that will serve as the key.
2. **Prepare the Plaintext**: Convert the plaintext into numbers and group them into blocks. If the last block is incomplete, add a filler character (e.g., "X").
3. **Matrix Multiplication**: Multiply the key matrix by the plaintext matrix and take the result modulo 26.
4. **Convert Back to Letters**: Convert the resulting numbers back to letters.

Example: Encrypting "HELLO" with a 2×2 Key Matrix

Step 1: Choose a Key Matrix

Let's use the following key matrix:

$$K = \begin{pmatrix} 6 & 24 \\ 1 & 13 \end{pmatrix}$$

Step 2: Prepare the Plaintext

Convert the plaintext "HELLO" into numbers:

- H = 7
- E = 4
- L = 11
- L = 11
- O = 14

Group into pairs (blocks of 2):

- Pair 1: HE → $\begin{pmatrix}7\\4\end{pmatrix}$
- Pair 2: LL → $\begin{pmatrix}11\\11\end{pmatrix}$
- Pair 3: O → Add "X" → $\begin{pmatrix}14\\23\end{pmatrix}$

The complete blocks are:

1. $\begin{pmatrix}7\\4\end{pmatrix}$ (HE)
2. $\begin{pmatrix}11\\11\end{pmatrix}$ (LL)
3. $\begin{pmatrix}14\\23\end{pmatrix}$ (OX)

Step 3: Encrypt Each Pair

Encrypt Pair HE:

$$\begin{pmatrix}6 & 24\\1 & 13\end{pmatrix}\begin{pmatrix}7\\4\end{pmatrix} = \begin{pmatrix}6\cdot 7 + 24\cdot 4\\1\cdot 7 + 13\cdot 4\end{pmatrix} = \begin{pmatrix}42+96\\7+52\end{pmatrix} = \begin{pmatrix}138\\59\end{pmatrix}$$

Now take modulo 26:

$$\begin{pmatrix}138 \mod 26\\59 \mod 26\end{pmatrix} = \begin{pmatrix}8\\7\end{pmatrix}$$

Convert back to letters:

- 8 → I
- 7 → H

Encrypted Pair: IH

Encrypt Pair LL:

$$\begin{pmatrix}6 & 24\\1 & 13\end{pmatrix}\begin{pmatrix}11\\11\end{pmatrix} = \begin{pmatrix}6\cdot 11 + 24\cdot 11\\1\cdot 11 + 13\cdot 11\end{pmatrix} = \begin{pmatrix}66+264\\11+143\end{pmatrix} = \begin{pmatrix}330\\154\end{pmatrix}$$

Now take modulo 26:

$$\begin{pmatrix} 330 \mod 26 \\ 154 \mod 26 \end{pmatrix} = \begin{pmatrix} 20 \\ 22 \end{pmatrix}$$

Convert back to letters:

- 20 → U
- 22 → W

Encrypted Pair: UW

Encrypt Pair OX:

$$\begin{pmatrix} 6 & 24 \\ 1 & 13 \end{pmatrix} \begin{pmatrix} 14 \\ 23 \end{pmatrix} = \begin{pmatrix} 6 \cdot 14 + 24 \cdot 23 \\ 1 \cdot 14 + 13 \cdot 23 \end{pmatrix} = \begin{pmatrix} 84 + 552 \\ 14 + 299 \end{pmatrix} = \begin{pmatrix} 636 \\ 313 \end{pmatrix}$$

Now take modulo 26:

$$\begin{pmatrix} 636 \mod 26 \\ 313 \mod 26 \end{pmatrix} = \begin{pmatrix} 10 \\ 25 \end{pmatrix}$$

Convert back to letters:

- 10 → K
- 25 → Z

Encrypted Pair: KZ

Step 4: Combine the Encrypted Letters

The complete encrypted text for "HELLO" is **IH UW KZ**.

Decryption Process

To decrypt the ciphertext, the following steps are taken:

1. **Find the Inverse of the Key Matrix**: Calculate the inverse of the key matrix KKK modulo 26.
2. **Convert the Ciphertext to Numbers**: Convert the ciphertext into numbers.
3. **Multiply by the Inverse Key Matrix**: Use matrix multiplication with the inverse key matrix and reduce modulo 26.
4. **Convert Back to Letters**: Convert the resulting numbers back to letters to retrieve the original plaintext.

Example: Decrypting "IH UW KZ"

1. **Inverse Key Matrix**:
   - The determinant d of the key matrix K is calculated as:

$$d = (6 \cdot 13 - 24 \cdot 1) = 78 - 24 = 54$$

   - Now find the modular inverse of $d \mod 26$. The modular inverse of 54 modulo 26 is 4 (since $54 \mod 26 = 2$ and the modular inverse of 2 mod 26 is 13).

   The inverse key matrix K^{-1} can be calculated as follows:

$$K^{-1} = \frac{1}{d} \begin{pmatrix} d_{22} & -d_{12} \\ -d_{21} & d_{11} \end{pmatrix} \mod 26$$

   After calculation:

$$K^{-1} \approx \begin{pmatrix} 1 & 24 \\ 1 & 22 \end{pmatrix} \mod 26$$

2. Convert Ciphertext to Numbers:

   - I = 8, H = 7, U = 20, W = 22, K = 10, Z = 25.
   - Group into pairs: (8, 7), (20, 22), (10, 25).

2. **Multiply by Inverse Key Matrix** and take modulo 26:
   - Decrypt (8, 7):

$$K^{-1} \begin{pmatrix} 8 \\ 7 \end{pmatrix} = \begin{pmatrix} 1 & 24 \\ 1 & 22 \end{pmatrix} \begin{pmatrix} 8 \\ 7 \end{pmatrix} \mod 26$$

   - Calculate and repeat for other pairs.
3. **Convert Back to Letters**: Convert the resulting numbers back to letters.

Summary

The Hill cipher effectively encrypts blocks of text using matrix multiplication, making it more secure than simple substitution ciphers. Understanding its encryption and decryption processes is essential for cryptography. If you need more examples or further explanations, feel free to ask!